

Navier-Stokes Vortex Dynamics

Numerical Formulation and Scientific Visualization Notes

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Abstract

This short technical note documents the mathematical and numerical formulation used in Version 1 of the *Navier-Stokes Vortex Dynamics* scientific visualization. The computational core is an actual two-dimensional pseudo-spectral integration of the incompressible Navier-Stokes equations in vorticity form on a periodic square. Velocity is recovered from the streamfunction in Fourier space, nonlinear transport is evaluated pseudo-spectrally, a 2/3 de-aliasing rule is applied, and time integration is performed with classical fourth-order Runge-Kutta. The animation additionally contains a conceptual three-dimensional vortex-filament sequence designed to illustrate vortex stretching. That final sequence is a visualization of the mechanism only; it is not a three-dimensional Navier-Stokes computation and does not constitute a numerical reproduction of a finite-time blow-up proof.

1 Governing equations

For an incompressible velocity field $\mathbf{u}(\mathbf{x}, t)$, the Navier-Stokes equations may be written as

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

where p is the kinematic pressure, $\nu > 0$ is the kinematic viscosity and $\mathbf{f}$ is an external forcing.

Introducing the vorticity

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad (2)$$

the three-dimensional vorticity equation contains the stretching term

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \Delta \boldsymbol{\omega} + \nabla \times \mathbf{f}. \quad (3)$$

The term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$ is central to the conceptual 3D finale of the visualization.

The numerical simulation itself is two-dimensional. In two dimensions, ω is a scalar normal to the plane and the stretching term is absent. The evolved equation is

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \Delta \omega + f_\omega. \quad (4)$$

2 Streamfunction formulation

On the periodic domain

$$\Omega = [0, 2\pi]^2,$$

a streamfunction ψ is introduced through

$$\Delta\psi = -\omega, \quad \mathbf{u} = (u, v) = (\psi_y, -\psi_x). \quad (5)$$

This construction satisfies incompressibility identically:

$$\nabla \cdot \mathbf{u} = \partial_x \psi_y - \partial_y \psi_x = 0.$$

Let $\widehat{\omega}_{\mathbf{k}}$ and $\widehat{\psi}_{\mathbf{k}}$ denote Fourier coefficients. Since

$$\widehat{\Delta\psi}_{\mathbf{k}} = -|\mathbf{k}|^2 \widehat{\psi}_{\mathbf{k}},$$

Equation (5) gives, for $\mathbf{k} \neq 0$,

$$\widehat{\psi}_{\mathbf{k}} = \frac{\widehat{\omega}_{\mathbf{k}}}{|\mathbf{k}|^2}, \quad \widehat{\psi}_{\mathbf{0}} = 0. \quad (6)$$

The velocity components are then reconstructed spectrally as

$$\widehat{u}_{\mathbf{k}} = ik_y \widehat{\psi}_{\mathbf{k}}, \quad \widehat{v}_{\mathbf{k}} = -ik_x \widehat{\psi}_{\mathbf{k}}. \quad (7)$$

3 Pseudo-spectral discretization

The domain is discretized on a uniform 128×128 periodic grid. Spatial derivatives are evaluated in Fourier space,

$$\widehat{\partial_x \omega} = ik_x \widehat{\omega}, \quad \widehat{\partial_y \omega} = ik_y \widehat{\omega}, \quad \widehat{\Delta \omega} = -|\mathbf{k}|^2 \widehat{\omega}. \quad (8)$$

The nonlinear advective term is evaluated in physical space:

$$\mathcal{N}(\omega) = u \omega_x + v \omega_y. \quad (9)$$

Thus the semi-discrete right-hand side used by the implementation is

$$\mathcal{R}(\omega, t) = -\mathcal{N}(\omega) + \nu \Delta \omega + f_\omega(\mathbf{x}, t). \quad (10)$$

To reduce aliasing generated by the quadratic nonlinearity, Fourier coefficients outside the 2/3 spectral region are removed. In the code this is implemented with the component-wise mask

$$|k_x| \leq N/3, \quad |k_y| \leq N/3.$$

4 Initial condition and forcing

The initial vorticity is a superposition of four localized Gaussian vortices and two smooth Fourier perturbations:

$$\begin{aligned} \omega_0(x, y) = & \sum_{j=1}^4 A_j \exp \left[-\frac{d_{\text{per}}(x, x_j)^2 + d_{\text{per}}(y, y_j)^2}{2\sigma_j^2} \right] \\ & + 0.30 \sin(3x + 2y) + 0.18 \cos(4x - y), \end{aligned} \quad (11)$$

followed by removal of the spatial mean. Here d_{per} denotes the shortest periodic displacement.

The vortex parameters are

j	x_j	y_j	A_j	σ_j
1	$0.30L$	$0.52L$	$+5.2$	0.34
2	$0.70L$	$0.48L$	-5.2	0.34
3	$0.50L$	$0.72L$	$+2.8$	0.27
4	$0.50L$	$0.28L$	-2.8	0.27

with $L = 2\pi$.

A weak, smooth time-dependent vorticity forcing is used:

$$f_\omega(x, y, t) = 0.12 \sin(2x + \phi) \cos(2y - 0.7\phi), \quad \phi = 0.40t. \quad (12)$$

5 Time integration

Time stepping uses the classical fourth-order Runge–Kutta scheme. Writing $\mathcal{R}$ for Equation (10),

$$k_1 = \mathcal{R}(\omega^n, t_n), \quad (13)$$

$$k_2 = \mathcal{R} \left(\omega^n + \frac{\Delta t}{2} k_1, t_n + \frac{\Delta t}{2} \right), \quad (14)$$

$$k_3 = \mathcal{R} \left(\omega^n + \frac{\Delta t}{2} k_2, t_n + \frac{\Delta t}{2} \right), \quad (15)$$

$$k_4 = \mathcal{R}(\omega^n + \Delta t k_3, t_n + \Delta t), \quad (16)$$

and

$$\omega^{n+1} = \omega^n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (17)$$

After every step, the mean vorticity is removed and the spectral de-aliasing mask is reapplied.

The principal numerical parameters are

Parameter	Value
Grid size	128×128
Periodic domain	$[0, 2\pi]^2$
Kinematic viscosity	$\nu = 2.5 \times 10^{-3}$
Time step	$\Delta t = 0.010$
Substeps per stored snapshot	4
Physics snapshots	10 s^{-1}
Visualization frame rate	30 fps
Animation duration	34 s
Tracer particles	1700

6 Diagnostics

Three scalar diagnostics are displayed during the simulation. The discrete kinetic-energy density is

$$E(t) = \frac{1}{2} \langle u^2 + v^2 \rangle, \quad (18)$$

the enstrophy density is

$$Z(t) = \frac{1}{2} \langle \omega^2 \rangle, \quad (19)$$

and the maximum resolved vorticity magnitude is

$$\|\omega\|_\infty = \max_{i,j} |\omega_{ij}|. \quad (20)$$

Here $\langle \cdot \rangle$ denotes the grid average. These are numerical diagnostics of the computed 2D field; they are not used as evidence of three-dimensional singular behavior.

7 Lagrangian tracers and visualization layer

A set of 1700 passive particles is advected by the numerically reconstructed velocity field. At particle position $\mathbf{x}_p = (x_p, y_p)$,

$$\frac{d\mathbf{x}_p}{dt} = \mathbf{u}(\mathbf{x}_p, t), \quad (21)$$

where velocity at off-grid positions is obtained by periodic bilinear interpolation. The tracers provide a Lagrangian visual complement to the Eulerian vorticity field and streamlines.

The rendering layer combines the magnitude $|\omega|$, vorticity contours, velocity streamlines, passive tracers, and the three live diagnostics. These elements are visual encodings of quantities computed from the 2D numerical solution.

8 Conceptual three-dimensional finale

The final spiral/filament sequence is deliberately separated from the two-dimensional solver. It is a parametric three-dimensional construction whose radius contracts while its axial scale increases,

visually evoking the geometric idea of vortex stretching. Its scientific role is to connect the 2D vorticity visualization to the stretching term in Equation (3).

This distinction is essential:

The 2D field is an actual numerical pseudo-spectral Navier-Stokes integration. The 3D filament is a conceptual vortex-stretching visualization, not a 3D Navier-Stokes solution and not a reproduction of a proof of finite-time blow-up.

9 Reproducibility and scope

The implementation uses NumPy FFTs for the spectral operations, Matplotlib for rendering and animation, and FFmpeg for MP4 export. A fixed random seed (2026) is used for reproducibility of the particle/background initialization.

This project is intended as a compact scientific-visualization experiment: the numerical layer exposes nonlinear transport, diffusion and vorticity evolution in a controlled two-dimensional periodic setting, while the conceptual finale provides a visual bridge to the genuinely three-dimensional vortex-stretching mechanism and the broader regularity question.

Repository note

Suggested repository contents:

- the V1 Python implementation;
- this technical note;
- the rendered MP4 or a link to the full visualization, available at [YouTube: Navier--Stokes in Motion](#);
- a concise README with execution requirements and the scientific distinction between the numerical 2D and conceptual 3D components.